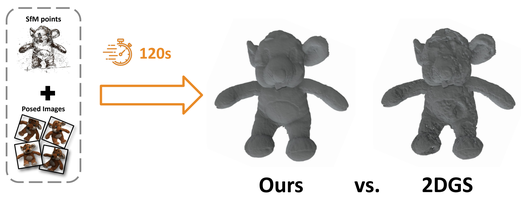
Instant Surface Reconstruction

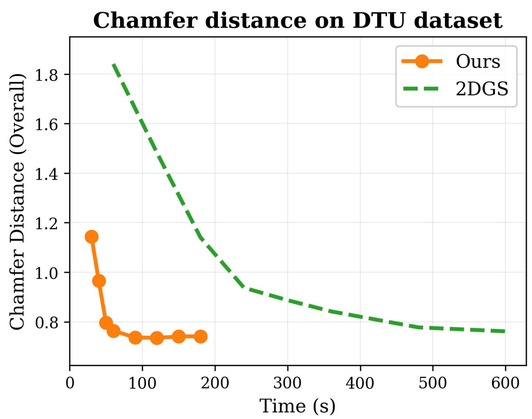
Lukas Haußmann, Jon Ayo Gieß Supervised by Lukas Höllein
TUM Visual Computing & Artificial Intelligence Lab

Motivation

- Accelerate high quality surface reconstruction from 2D Gaussian Splatting through **dense initialization.**
- 2DGS's sparse initialization requires long iterative optimization → slow.



Quantitative Results



Method	24	37	40	55	63	65	69	83	Time
3DGS	2.14	1.53	2.08	1.68	3.49	2.21	1.43	2.07	15min
SuGaR	1.47	1.33	1.13	0.61	2.25	1.71	1.15	1.63	1h
2DGS	**0.46**	**0.80**	**0.33**	**0.37**	0.95	0.86	0.80	1.25	15min
2DGS*	1.38	1.26	1.41	0.93	2.63	2.14	1.25	1.71	2min
Ours	0.50	0.89	0.39	0.39	**0.89**	**0.79**	**0.74**	**1.19**	2min

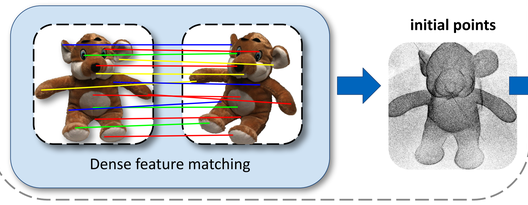
Method	97	105	106	110	114	118	122	Mean	Time
3DGS	2.22	1.75	1.79	2.55	1.53	1.52	1.50	1.97	15min
SuGaR	1.62	1.07	0.79	2.45	0.98	0.88	0.79	1.32	1h
2DGS	**1.24**	0.67	0.67	1.24	**0.39**	0.64	**0.47**	0.74	15min
2DGS*	1.66	1.30	1.12	2.07	1.22	1.06	1.09	1.48	2min
Ours	1.36	**0.64**	**0.59**	**1.12**	0.44	**0.61**	0.49	**0.74**	2min

Methodology

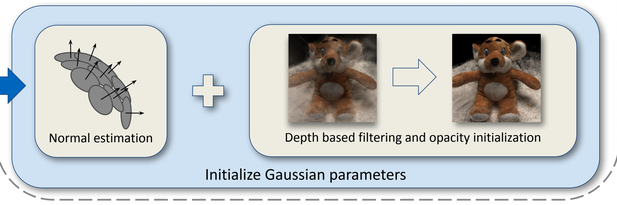
Input: RGB posed images, SfM point cloud



Generate points: Use EDGS [1] method: dense feature matching + camera poses to generate a very dense point cloud



Gaussian initialization: Estimate surface normals, filter points based on estimated depth, assign gaussian opacities based on local point density
→ reduce penalty for inaccurate initial points



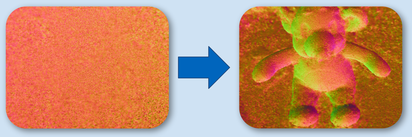
Ablations

Chamfer distance ↓ on DTU dataset at 120s.

Experiment	Accuracy	Completeness	Overall
w/o normal estimation	0.748	0.812	0.780
with density control	0.689	0.783	0.736
w/o density control	0.702	0.772	0.737
w/o depth based filter/opacity init	0.752	0.788	0.770
full model	0.689	0.783	**0.736**

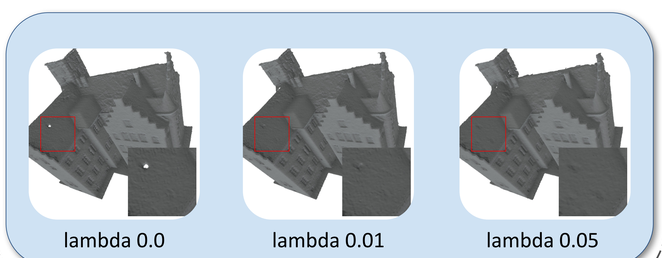
Densification is **not required**, can improve robustness against too sparse regions

Normal estimation: Rendered normal maps

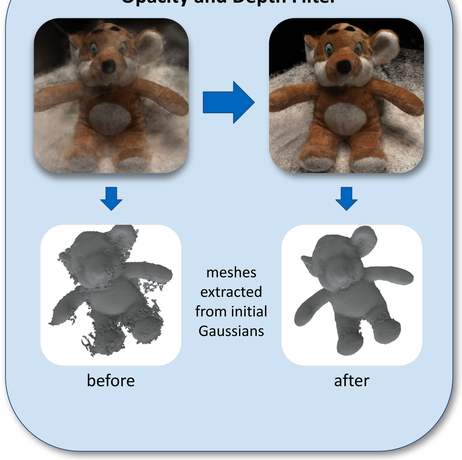


Normal consistency regularization

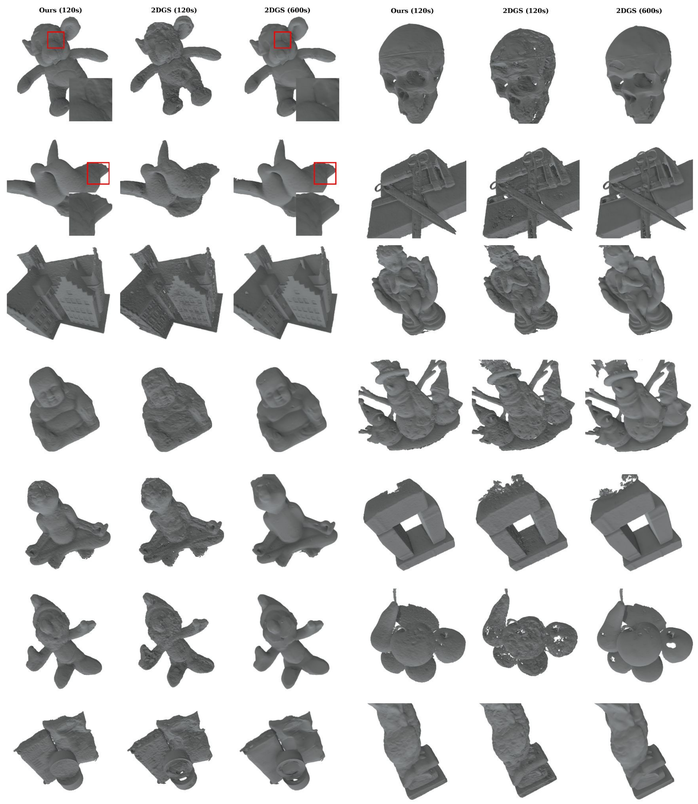
Experiment	Accuracy	Completeness	Overall
lambda 0.0	0.737	**0.770**	0.754
lambda 0.01	**0.700**	0.770	**0.735**
lambda 0.05	0.719	0.829	0.774



Opacity and Depth Filter



Qualitative Results



Contributions

- We introduce a dense 2D Gaussian Splatting initialization for more efficient surface reconstruction.
- We leverage **dense feature matching, normal** and **depth estimation** to achieve accurate set of initial Gaussians.
- Our method achieves faster convergence than existing methods.

[1] Kotovenko et al., "EDGS", arXiv:2504.13204 (2025)